uppermost part of the figure. John Wheeler coined the term quantum foam to describe the frenzy revealed by such an ultramicroscopic examination of space (and time)—it describes an unfamiliar arena of the universe in which the conventional notions of left and right, back and forth, up and down (and even of before and after) lose their meaning. It is on such short distance scales that we encounter the fundamental incompatibility between general relativity and quantum mechanics. The notion of a smooth spatial geometry, the central principle of general relativity, is destroyed by the violent fluctuations of the quantum world on short distance scales. On ultramicroscopic scales, the central feature of quantum mechanics—the uncertainty principle—is in direct conflict with the central feature of general relativity—the smooth geometrical model of space (and of spacetime).

In practice, this conflict rears its head in a very concrete manner. Calculations that merge the equations of general relativity and those of quantum mechanics typically yield one and the same ridiculous answer: infinity. Like a sharp rap on the wrist from an old-time schoolteacher, an infinite answer is nature's way of telling us that we are doing something that is quite wrong.6 The equations of general relativity cannot handle the roiling frenzy of quantum foam.

Notice, however, that as we recede to more ordinary distances (following the sequence of drawings in Figure 5.1 in reverse), the random, violent small-scale undulations cancel each other out—in much the same way that, on average, our compulsive borrower's bank account shows no evidence of his compulsion—and the concept of a smooth geometry for the fabric of the universe once again becomes accurate. It's like what you experience when you look at a dot-matrix picture: From far away the dots that compose the picture blend together and create the impression of a smooth image whose variations in lightness seamlessly and gently change from one area to another. When you inspect the picture on finer distance scales you realize, however, that it markedly differs from its smooth, long-distance appearance. It is nothing but a collection of discrete dots, each quite separate from the others. But note that you become aware of the discrete nature of the picture only when you examine it on the smallest of scales; from far away it looks smooth. Similarly, the fabric of spacetime appears to be smooth except when examined with